MEMORANDUM

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RE: **Pipe Flow Analysis to Determine Flow Type and Measure Volumetric Flowrate**

In this experiment, the flowrate through a tube and its uncertainty was measured and compared to model results for various water heights. The transition from laminar to turbulent flow was also visually examined using colored dye. The experimental setup included a tank with columns of various heights, a tube with a rounded entrance, and a bucket used to measure the flowrate. Understanding fluid behavior in pipes is important because it is prevalent in many real-life situations such as plumbing systems or blood flowing through medical devices. The calculated experimental flowrate for the highest tank height was 0.717 gpm ± 0.083%. For the lowest tank height tested, 0.75 feet, the calculated experimental flowrate was 0.54 gpm ± 0.13%, indicating that lower tank heights will lead to lower flowrates. The model yielded a flowrate of 0.7 gpm ± 5.7% for the highest tank height, indicating that the model and experimental results agree.

Background

It is important to understand fluid flow through pipes. The mass flowrate through a pipe can be experimentally measured by collecting and weighing the fluid output from the pipe while recording the time that the water was collected. To obtain the volumetric flowrate, the density of the fluid must be known. This simple procedure is useful and allows for the calculation of volumetric flowrate, a useful parameter for fluid analysis.

Additionally, the type of flow, laminar or turbulent, helps to characterize the fluid flow. The flow is considered turbulent when the inertial forces are more dominant over viscous forces. Conversely, laminar flow is when the viscous forces are more dominant than the inertial forces. As a result, laminar flow is associated with high amounts of friction, and the fluid flows at low velocities. The type of flow can be correlated with the Reynold's number; a Reynolds number of around generally 2300 indicates a transition from laminar to turbulent flow.

Procedure/Analysis

The experimental setup for the lab consisted of a large tank with columns of four different heights, a tube exiting the tank, and a pump used to fill the tank with water. At the exit of the tube, a bucket was used to collect the water flowing out of the tube. Additionally, a scale and stopwatch were used for data collection of the weight of the bucket and time required to fill the bucket, respectively. Figure 1 shows the setup for the pipe flow experiment.



Figure 1. Experimental setup used to measure the volumetric flowrate and observe the laminar or turbulent characteristics of the flow [1].

The empty bucket was weighed prior to starting the experiment. The tank was filled with water using a pump by opening the bypass valve until it reached a height above the tallest column in the tank. The starting height of the water was recorded using a ruler and the resolution uncertainty of this measurement was noted. The outlet to the tube with the rounded entrance was opened to allow water to flow from the tank to the bucket. Water was collected in the bucket for approximately 4 minutes. The water in the bucket was weighed using a scale, and the resolution uncertainty of this measurement was recorded.

For each column height, the overflow valve was opened, and the tank reached a new height. The procedure outlined above was repeated for each height. After the data for each height was recorded, dye was injected into the water to observe how the transition from turbulent to laminar flow.

To calculate the flowrate from the experimental setup, Equation 1 was used,

$$Q = \frac{W}{\rho g t} \tag{1}$$

where Q is the volumetric flowrate, W is the weight of the water, ρ is the density of the water, g is the acceleration of gravity, and t is the amount of time that the water was collected. This process was repeated for all of the tank heights tested.

To compare the experimental results to a model for incompressible, steady flow through a pipe with losses, Equation 2 was used,

$$Q = \bar{V}A = \sqrt{\frac{2gh}{\alpha + K_L + f\left(\frac{l}{D}\right)}}\left(\frac{\pi}{4}D^2\right) \tag{2}$$

where $\bar{V}$ is the average velocity of the fluid, h is the tank height, f is the friction factor, l is the length of the pipe, D is the diameter of the pipe, g is the acceleration of gravity, α is the kinetic energy coefficient (assumed to be 1 for turbulent flow), and K_L is a loss coefficient. To find the appropriate flowrate, the friction factor was iterated by guessing a friction factor and recalculating based on the Reynolds number. The Reynolds number was calculated using Equation 3,

$$Re_D = \frac{\rho \bar{V} D}{\mu} \tag{3}$$

where ρ is the density of water and μ is the dynamic viscosity of the fluid. Once the friction factor converged to a value, it was used in Equation 2 to find the theoretical flowrate. This was repeated for each tank height.

After calculating the flowrates, the uncertainty for the experimental values and theoretical values were calculated. The uncertainty for the experimental flowrate was calculated using Equation 4,

$$\frac{u_Q}{Q}_{exp} = \sqrt{\left(\frac{u_W}{W}\right)^2 + \left(\frac{u_t}{t}\right)^2} \tag{4}$$

where u_W is the resolution uncertainty in the scale used to weigh the water and u_t is the resolution uncertainty for the stopwatch that recorded the time that the water was collected.

For calculating our model flowrate uncertainty using our model, Equation 5 was used,

$$\frac{u_Q}{Q} = \sqrt{\left(\frac{\partial Q}{\partial h}u_h\right)^2 + \left(\frac{\partial Q}{\partial f}u_f\right)^2 + \left(\frac{\partial Q}{\partial K_L}u_{K_L}\right)^2} \tag{5}$$

where ∂Q denotes the partial derivative of Equation 2 with respect to the appropriate variable, u_h is the uncertainty in the tank height measurement, u_f is the uncertainty in the friction factor, and u_{K_L} is the uncertainty in the loss coefficient.

To determine the range of Reynolds numbers for the transition from turbulent to laminar flow, the volumetric flowrate was calculated for an assumed friction factor. The velocity was calculated from the volumetric flowrate using Equation 2. Similar to the calculations for the model flowrate, the friction factor was initially assumed, so it must be iterated upon to find the actual Reynolds number. As a result, a new

friction factor was found using the previous Reynolds number and the Moody diagram. The process was repeated until the friction factor converged to a steady value and the volumetric flowrate and Reynolds number were calculated using this converged value and Equation 3.

Results & Discussion

Table 1 summarizes the results for the experimental flowrate and shows percent uncertainty derived from the calculations explained below. The weight and time for the water collected are also shown in Table 1, which were used to calculate the experimental flowrate. Sample calculations for the highest tank height for the experimental and theoretical model flowrates are shown in Attachment A.

Table 1. Calculated flowrate and percent uncertainty corresponding to the flowrate obtained experimentally

Tank Height, h(ft)	Experimental Flowrate, Q (gpm)	Experimental % Uncertainty	Weight, W (lbf)	Time (s)
1.26	0.717	0.083%	24.20	242.64
1.10	0.66	0.10%	19.58	212.69
0.91	0.60	0.11%	17.58	210.3
0.75	0.54	0.13%	15.68	210.57

The calculated flowrates, Reynolds number, friction factor, and uncertainty for the flowrates obtained from the model are shown in Table 2. Attachment B shows all of the friction factors and Reynolds numbers used for the iterative process. Attachment C includes sample calculations for the iterative process for the friction factor used to calculate the model flowrates.

Table 2. Calculated flowrates, Reynolds number, friction factor, and uncertainty for model for steady, incompressible flow

Tank height, h (ft)	Reynolds Number	Friction Factor	Flow Rate, Q (gpm)	Model % Uncertainty
1.26	5300	0.037	0.705	5.5%
1.10	4900	0.038	0.654	5.4%
0.91	4450	0.039	0.588	5.4%
0.75	3950	0.040	0.5277	5.4%

The model and experimental flowrates are relatively consistent. These values and their uncertainty are compared later in the memo. The experimental data and the model both have a relatively low percent uncertainty, but the percent uncertainty is much lower than the model uncertainty. This is because the model has a very large uncertainty for the loss coefficient and the resolution uncertainties of the values used for the experimental uncertainty are very small.

The experimental flowrate uncertainty was calculated using Equation 4. As a sample calculation for the first tabulated percent error, values for each variable can be plugged in:

$$\frac{u_Q}{Q}_{exp} = \sqrt{\left(\frac{0.02\ lbf}{24.20\ lbf}\right)^2 + \left(\frac{0.01\ s}{242.62\ s}\right)^2} = 0.000827473 = 0.083\%$$

For calculating our model flowrate uncertainty using our model, Equation 5 was used.
Using the given specifications for the individual uncertainties ($u_h = 0.0625\ ft$, $\frac{u_f}{f} = 10\%$, $\frac{u_{K_L}}{K_L} = 70\%$) [1]:

$$\frac{u_Q}{Q}_{mod} = \sqrt{\left(\frac{Q}{2h}\right)^2 + \left(-\frac{1}{2} * \frac{\left(\frac{L}{D}\right)}{\alpha + K_L + \left(\frac{fL}{D}\right)} * (0.10u_h)\right)^2 + \left(-\frac{1}{2} * \frac{Q}{\alpha + K_L + \left(\frac{fL}{D}\right)} * (0.70u_{K_L})\right)^2}$$

For a sample calculation showing the first tabulated percent error from the model, the values for each variable can be substituted into the equation above. First, we will calculate the denominator for the second two terms in the square root:

$$den = \alpha + K_L + \left(\frac{fL}{D}\right) = 1 + 0.7 + \left(0.037 * \frac{2.467\ ft}{0.022\ ft}\right) = 5.849$$

Next, plug this value in for the denominator, and plug in all other variables:

$$\frac{u_Q}{Q}_{mod} = \sqrt{\left(\frac{0.0016\frac{ft^3}{s}}{2 * 1.26\ ft}\right)^2 + \left(-\frac{1}{2} * \frac{\left(\frac{2.467\ ft}{0.022\ ft}\right)}{5.849} * (0.10)\right)^2 + \left(-\frac{1}{2} * \frac{0.0016\frac{ft^3}{s}}{5.849} * (0.70)\right)^2}$$

$$\frac{u_Q}{Q}_{mod} = 0.054889 = 5.49\%$$

For all four trials, the experimental flowrate uncertainty was approximately 0.08-0.13%. From the two uncertainty terms that make up the total uncertainty, the weight term was much larger than the time term, corresponding to its greater impact on the total uncertainty than the time term. The results for the calculated uncertainty for all tank heights and for the experimental and theoretical model are shown in Appendix A. As shown in Table 2, these resolution-based terms were gathered from the scale and the stopwatch used to conduct the experiment.

The model flowrate uncertainty was approximately 5.4% for all four trials and was dominated by uncertainty in the entrance loss coefficient K_L, which was given by the lab manual as 70% (or 0.7). The uncertainty in the friction factor (10% or 0.1 as stated in the lab manual) also contributed a significant amount, sitting at values around 3.5-3.6%. The model height uncertainty was calculated to be around 0.06-0.08%, which attributed negligible effects to the overall total flowrate uncertainty for the model.

Figure 2 shows the flowrate as a function of tank height for both the experimental measurements (symbols) and the model prediction (line). The uncertainty for the experimental data and theoretical model are shown

in Figure 2 with error bars indicating the uncertainty for the flowrate. It is important to note that the experimental data uncertainty was very small (around 0.0007 gpm) and the model uncertainty was much larger (around 0.060 gpm), so the error bars for the model overlap completely with the error bars for the experimental data.

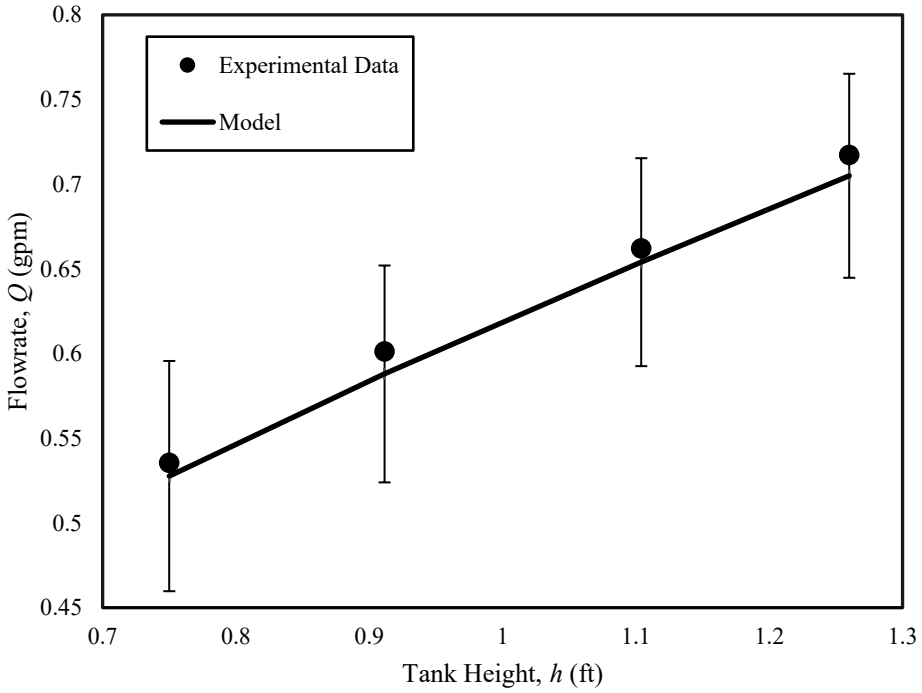


Figure 2. Experimental and Model Results for Flowrate versus Tank Height

In general, the experimental data follows the same increasing trend as the model, with flowrate increasing as the tank height increases. This trend makes physical sense because a higher water level in the tank provides a larger hydrostatic head to drive the flow. As the available head increases, the pressure at the tube entrance increases, resulting in a higher velocity and therefore a higher volumetric flowrate. When the water level is lower, the driving head is reduced and the flowrate decreases. The small differences between the experimental and model flowrates are expected and are likely due to simplifying assumptions in the model, such as assuming fully developed one-dimensional flow and using idealized loss coefficients. Overall, the model captures the main behavior of the system reasonably well. Additionally, the uncertainty ranges for the experimental data and the model overlap, indicating agreement between the values.

While the tank drained, the transition from turbulent to laminar flow was observed using colored dye injected into the tank. While the tank drained, the dye was originally mixed thoroughly into the water, indicating turbulent flow because turbulent flow is characterized by a high degree of mixing and random motion. For turbulent flow, the fluid has eddies and the fluid is very irregular. However, laminar flow is organized, and the fluid moves in parallel lines. This was observed in the experiment when the colored dye traveled parallel to the water, and there was no mixing between the dye and water.

While observing the transition to laminar flow, the transition started to occur at a tank height of 8.5 feet and the flow fully transitioned to laminar flow at a tank height of 7.3 feet. The recorded tank heights were used to find the range of Reynolds numbers where the flow transitioned. The Reynolds number was calculated by iterating upon the flowrate and friction factor until a converged value was reached. The results for the Reynolds number while the flow transitioned to laminar flow are shown in Table 3.

Table 3. Reynolds numbers for transition to laminar flow

	Initial Signs of Turbulent Flow	Full Transition to Laminar Flow
Tank Height (ft)	8.5	7.3
Reynolds Number	6060	5426

The range of Reynolds numbers for the transition to laminar flow is approximately 5400 to 6060. This is much higher than the expected transition Reynolds number of 2300. This could mean that laminar flow is harder to achieve than would be typically expected because of the experimental conditions. For example, the pipe roughness and frictional losses in the pipe may be greater than assumed, which would cause the transition to laminar flow to be later.

References

[1] ME 347 Lab 4 Manual, Arturo Cajal.

List of Attachments

A. Sample Calculations for Experimental and Model Flowrates
B. Model Flowrate Friction Factors, Reynolds Number, and Flowrates
C. Iterations for Model Calculations

ENGINEERING DATA SHEET	Sheet No.	A – 1/1	Date:
Attachment A: Sample Calculations for Experimental and Model Flowrates	Prepared	GL	02/07/26
	Checked	KG	02/08/26
	Approved	LMS	02/08/26

The calculations for the experimental flowrates were completed using the weight of the water in the bucket, the density of the water, the acceleration due to gravity, and the amount of time that the water in the bucket was collected. For the highest tank height, 1.26 ft, the experimental flowrate was computed,

$$Q = \frac{24.20\ lbf}{62.4\ lbm/ft^3\left(32.174\frac{lbf}{fts^2}\right)242.64\ s}\frac{448\ gpm}{1\ ft^3/s} = 0.717\ gpm$$

To calculate the model flowrate, the model for steady, incompressible flow was used,

$$Q = \bar{V}A = \sqrt{\frac{2(32.174\ \frac{lbf}{fts^2})}{1 + 0.04 + 0.033\left(\frac{2.467ft}{0.022\ ft}\right)}}\left(\frac{\pi}{4}(0.022ft)^2\right) = 0.705\ gpm$$

ENGINEERING DATA SHEET	Sheet No.	B – 1/1	Date:
Attachment B: Model Flowrate Friction Factors, Reynolds Number, and Flowrates	Prepared	LMS	02/07/26
	Checked	KG	02/08/26
	Approved	GL	02/08/26

For the model flowrates the friction factor was iterated and converged to a single value. The iterated friction factors are shown in Tables 4, 5, 6, and 7. A sample calculation of this iterative process is shown below.

Tables 4,5,6,7. Model Predictions for Reynolds Number and Friction Factor

h=1.26ft Model						
Iteration	f_guess	Q(ft^3/s)	Q(gpm)	V (Q/A)	Re	k_new
Iteration 1	0.02000	0.00188849	0.84760998	4.96970473	9939.40947	0.031688109
Iteration 2	0.03142	0.00160173	0.71890608	4.21508837	8430.17673	0.033020018
Iteration 3	0.03285	0.00157431	0.70659827	4.14292522	8285.85044	0.033162877
Iteration 4	0.03300	0.00157152	0.70534355	4.13556858	8271.13716	0.033177615
Iteration 5	0.03302	0.00157114	0.70517676	4.13459065	8269.1813	0.033179577

h=1.10ft Model						
Iteration	f_guess	Q(ft^3/s)	Q(gpm)	V (Q/A)	Re	k_new
Iteration 1	0.02000	0.00176772	0.79340572	4.65189444	9303.78887	0.032215992
Iteration 2	0.03221	0.00148496	0.6664941	3.90778659	7815.57317	0.033650868
Iteration 3	0.03365	0.00145984	0.65521911	3.84167905	7683.35809	0.033794709
Iteration 4	0.03379	0.00145746	0.65415319	3.83542938	7670.85876	0.033808467
Iteration 5	0.03380	0.00145729	0.65407725	3.83498414	7669.96828	0.033809448

h=.91ft Model						
Iteration	f_guess	Q(ft^3/s)	Q(gpm)	V (Q/A)	Re	k_new
Iteration 1	0.02000	0.00160618	0.72090363	4.22680039	8453.60078	0.03299712
Iteration 2	0.03299	0.00133675	0.5999751	3.51777252	7035.54503	0.034547135
Iteration 3	0.03454	0.0013129	0.58926697	3.45498861	6909.97723	0.034703024
Iteration 4	0.03470	0.00131051	0.58819396	3.44869739	6897.39478	0.03471884
Iteration 5	0.03471	0.00131036	0.5881271	3.44830533	6896.61066	0.034719827

h=0.75ft Model						
Iteration	f_guess	Q(ft^3/s)	Q(gpm)	V (Q/A)	Re	k_new
Iteration 1	0.02000	0.001457	0.6539453	3.83421044	7668.42089	0.033811154
Iteration 2	0.03381	0.001201	0.53904464	3.16052522	6321.05045	0.035484538

Iteration 3	0.03548	0.00117838	0.5288918	3.10099712	6201.99425	0.03565362
Iteration 4	0.03565	0.00117615	0.52789015	3.09512423	6190.24846	0.035670521
Iteration 5	0.03567	0.00117589	0.52777268	3.09443549	6188.87099	0.035672506

ENGINEERING DATA SHEET	Sheet No.	C – 1/1	Date:
Attachment C: Iterations for Model Calculations	Prepared	GC	02/07/26
	Checked	LMS	02/08/26
	Approved	GL	02/08/26

ITERATION SAMPLE CALCULATION

MODEL ASSUMPTIONS

$D = 0.264\ in = 0.0220\ ft$

$L = 29.6\ in = 2.4667\ ft$

$r = 0.6625\ in$

$h = 1.26\ ft$

$g = 32.2\ ft/s^2$

1) ENTRANCE LOSS COEFF

$\frac{r}{D} = \frac{0.625}{0.264} = 0.237 \geq 0.15$

(TABLE 8.2)

$K_L = 0.04$

Table 8.2
Minor Loss Coefficients for Pipe Entrances

Entrance Type	Minor Loss Coefficient, K^a
Reentrant	0.5 – 1.0 (depending on length of pipe entrance)
Square-edged	0.5
Rounded	r/D: 0.02 → K: 0.3; 0.06 → 0.2; ≥0.15 → 0.04

[a] Based on $h_L = K(\bar{V}^2/2)$, where $\bar{V}$ is the mean velocity in the pipe.
Source: Data from Reference [12].

2) GEOMETRY

$A = \frac{\pi}{4} D^2 = \frac{\pi}{4}(0.0220)^2 = 3.80 \times 10^{-4}\ ft^2$

$\frac{L}{D} = \frac{2.4667}{0.0220} = 112.1$

3) PICK α_2

FOR TURBULENT / TYPICAL PIPE FLOW

$\alpha_2 \approx 1$

4) WATER PROPERTIES

$\nu \approx 1.1 \times 10^{-5}\ ft^2/s$

5) ITERATION FOR $h = 1.26\ ft$

$$Q = A\sqrt{\frac{2gh}{\alpha_2 + K_L + f\left(\frac{L}{D}\right)}}$$

ITER	f_{USED}	$Q\ (ft^3/s)$	$Q\ (gpm)$	Re	f_{new}
1	0.02	0.00184	0.84761	9939	0.0316
2	0.03142				
3	0.03283				
4	0.03300	0.00157	0.705		
5	0.03302	0.00157	0.705	8264	0.03318

ITERATION 1:

$$Q = 3.8 \times 10^{-4}\ ft^2 \sqrt{\frac{2\,(32.174\ ft/s^2)\,(1.26\ ft)}{1 + 0.02\,(112.1)}} = 0.001884\ \frac{ft^3}{s} = 0.848\ gpm$$

CONVERGED FINAL VALUES AFTER ITERATION

$$V = \frac{Q}{A} = \frac{0.001884\ ft^3/s}{3.80 \times 10^{-4}\ ft^2} = 4.9697\ \frac{ft}{s}$$

$$Re = \frac{VD}{\nu} = \frac{4.9697\ ft/s \cdot 0.0220\ ft}{1.1 \times 10^{-5}\ ft^2/s} = 9939$$

BLASIUS $f_{new} = \frac{0.3164}{Re^{0.25}} = \frac{0.3164}{9939^{0.25}} = 0.0316$